

Scalable Data Science
NPTEL 2026
Assignment-4

1. A stream of length m is summarized using a SpaceSaving sketch with a single counter. Which of the following frequencies is sufficient to ensure that an element is present in the sketch at the end of the stream, irrespective of the order of arrival?
 - A. More than $m/4$
 - B. More than $m/2$
 - C. More than $m/3$
 - D. Any positive frequency, provided the element appears near the end.

Solution Answer:(B) With a single counter ($k = 1$), the SpaceSaving algorithm guarantees

$$f_x - \frac{m}{2} \leq \hat{f}_x \leq f_x.$$

Therefore, if an element x has frequency $f_x > m/2$, then

$$\hat{f}_x \geq f_x - \frac{m}{2} > 0,$$

so x must occupy the counter at the end of the stream, regardless of the order of arrival.

2. Let H_d denote the $d \times d$ Hadamard matrix, and let $x \in \mathbb{R}^d$. Consider the transformed vector $y = H_d x$. Which of the following statements is necessarily true?
 - A. The vector y is always dense, regardless of x .
 - B. If x is dense, then y is sparse.
 - C. Multiplication by H_d alone leaves x unchanged; a random diagonal matrix D is required for any effect.
 - D. No general conclusion about the sparsity pattern of y can be made.

Solution Answer:(D) Multiplication by a Hadamard matrix mixes the coordinates of x , but whether the result is sparse or dense depends on the structure of x .

3. A data stream of length 280 is processed using the Misra-Gries algorithm with $k = 7$ counters. What is the minimum frequency that guarantees an element will still appear among the stored candidates after the entire stream has been processed?
 - A. 40

- B. 7
- C. 41
- D. 14

Solution Answer:(C) In the Misra-Gries algorithm, any element with frequency strictly greater than m/k is guaranteed to be present at the end; since $m/k = 280/7 = 40$, the required frequency is 41.

4. A data-dependent hashing scheme partitions points according to the signs of their projections onto eigenvectors of the data covariance matrix. Which statement best describes the success probability of finding a near neighbor?
- A. It is always at least 80%.
 - B. It depends on the spectral properties of the dataset.
 - C. It is zero because the partitioning is deterministic.
 - D. It is always at least 60%.

Solution Answer:(B) The success probability of a data-dependent LSH based on eigenvector projections depends on the spectral properties of the dataset. There is no universal guarantee such as 60% or 80%, and the fact that the hash is deterministic does not imply zero success probability.

5. Suppose you are running the Space Saving algorithm on an arrival-only stream of length $m = 10,000$ with error tolerance $\epsilon = 0.02$, so $k = 1/\epsilon = 50$. At the end of the stream, an item X has estimated frequency $f(X) = 450$.

According to the guarantees of the Space Saving algorithm, what is the tightest interval for the true frequency $f(X)$?

- (a) A. $[250, 450]$
- (b) B. $[450, 650]$
- (c) C. $[250, 650]$
- (d) D. $[0, 450]$

Solution Answer:(A) Space Saving guarantee:

$$f(X) - \epsilon m \leq f(X) \leq f(X)$$

Compute error bound:

$$\epsilon m = 0.02 \times 10,000 = 200$$

Substitute values:

$$450 - 200 \leq f(X) \leq 450$$

$$250 \leq f(X) \leq 450$$

6. A Count-Min sketch is used to summarize a stream. After processing the stream, an element x that actually appeared 9 times is queried. The sketch reports an estimated frequency of 7. Which of the following conclusions is correct?

- A. Such an underestimate is possible due to hash collisions; additional queries are needed to verify the result.
- B. The implementation is incorrect, since a Count-Min sketch never returns a value smaller than the true frequency.

Solution Answer:(B) A Count-Min sketch always overestimates or exactly matches the true frequency because collisions only increase counts; therefore an estimate smaller than the true count indicates a faulty implementation.

7. Let A be a matrix with m rows and n columns. Your goal is to pick a subset of columns of A . Your friend gives you a matrix S and says that if you multiply A with S (i.e. you calculate AS) you will get a subset of c columns of matrix A . What do you think the structure of matrix S should be? You do not want any scaling of the columns.

- A. S should be an orthonormal matrix.
- B. S is a matrix with n rows and c columns and each column of S has only one entry as 1 and all other entries in the column are 0.
- C. S is a matrix with n rows and c columns and each row of S has only one entry as 1 and all other entries in the row are 0.
- D. S is a matrix with n rows and c columns and each column of S has only c entries, each as $1/\sqrt{c}$ and all other entries in the column are 0.

Solution Answer:(B) S is a matrix with n rows and c columns and each column of S has only one entry as 1 and all other entries in the column are 0.

8. Let $x \in \mathbb{R}^d$ be a sparse vector, and let $nnz(x)$ denote the number of its nonzero entries. Suppose $S \in \mathbb{R}^{n \times d}$ is a sparse matrix in which each column contains exactly c nonzero entries. What is the tightest asymptotic upper bound on the time required to compute the matrix-vector product Sx ?

- A. $\Theta(nd)$
- B. $O(c \cdot nnz(x))$
- C. $O(n \cdot nnz(x))$
- D. $O(nnz(x))$

Solution Answer:(B) Since only the $nnz(x)$ nonzero coordinates contribute and each corresponding column of S contains c nonzero entries, the product Sx can be computed in $O(c \cdot nnz(x))$ time.

9. During the derivation of the Count-Min Sketch upper bound over an arrival-only stream, let Y_i be the value stored in slot $h_i(x)$ of row i . If $f(x)$ is the true frequency of item x , total stream length is m , and the number of columns is $d = \frac{2}{\epsilon}$, what is the expected value of the noise due to hash collisions with all elements $y \neq x$?

- (a) A. ϵm

- (b) B. $\frac{\epsilon}{2}(m - f(x))$
- (c) C. $\delta \cdot (m - f(x))$
- (d) D. $\frac{\epsilon m}{2} + f(x)$

Solution Answer:(B) In Count-Min Sketch,

$$Y_i = f(x) + \sum_{y \neq x, h_i(y)=h_i(x)} f(y)$$

Taking expectation:

$$\mathbb{E}[Y_i - f(x)] = \sum_{y \neq x} f(y) \cdot \Pr(h_i(y) = h_i(x))$$

Since hashing is uniform over d buckets:

$$\Pr(h_i(y) = h_i(x)) = \frac{1}{d}$$

Given $d = \frac{2}{\epsilon}$:

$$\frac{1}{d} = \frac{\epsilon}{2}$$

Also,

$$\sum_{y \neq x} f(y) = m - f(x)$$

Therefore,

$$\mathbb{E}[Y_i - f(x)] = \frac{\epsilon}{2}(m - f(x))$$

10. Suppose a random projection matrix $S \in \mathbb{R}^{s \times n}$ is used to map vectors from $\mathbb{R}^n$ to $\mathbb{R}^s$, where

$$s = O\left(\frac{1}{\epsilon^2} \log \frac{1}{\delta}\right).$$

Assume that the only guarantee available is

$$\mathbb{E}[\|Sx\|_2^2] = \|x\|_2^2$$

for every vector $x \in \mathbb{R}^n$, and no information about the variance of $\|Sx\|_2^2$ is known. Which of the following statements can be concluded?

- A. For every fixed x , $\|Sx\|_2^2$ is within a $(1 \pm \epsilon)$ factor of $\|x\|_2^2$ with probability at least $1 - \delta$.
- B. The above guarantee is valid only when $\delta = \frac{1}{2}$.
- C. For any fixed x and any $\alpha > 0$,

$$\Pr\left[\|Sx\|_2^2 \geq \frac{\|x\|_2^2}{\alpha}\right] \leq \alpha.$$

Solution Answer:(C) Knowing only that

$$\mathbb{E} [\|Sx\|_2^2] = \|x\|_2^2$$

does not imply concentration around the mean, so we cannot conclude a $(1 \pm \epsilon)$ relative approximation with probability $1 - \delta$. However, by Markov's inequality, for any $\alpha > 0$,

$$\Pr \left[\|Sx\|_2^2 \geq \frac{\|x\|_2^2}{\alpha} \right] \leq \frac{\mathbb{E} [\|Sx\|_2^2]}{\|x\|_2^2/\alpha} = \alpha.$$

11. To reduce dimensionality, a data scientist multiplies unit vectors in $\mathbb{R}^d$ by a random Gaussian matrix $A \in \mathbb{R}^{k \times d}$ whose entries are i.i.d. $N(0, 1)$. If $\|x\|_2 = 1$, what is the expected value of $\|Ax\|_2^2$?

- A. 1
- B. k
- C. d
- D. $1/k$

Solution Answer:(B) Let $a_1, a_2, \dots, a_k$ denote the rows of A . Then

$$\|Ax\|_2^2 = \sum_{i=1}^k (a_i^T x)^2.$$

Since the entries of A are i.i.d. $N(0, 1)$ and $\|x\|_2 = 1$, each $a_i^T x$ is distributed as $N(0, 1)$. Hence,

$$\mathbb{E} [(a_i^T x)^2] = 1,$$

and therefore

$$\mathbb{E} [\|Ax\|_2^2] = \sum_{i=1}^k \mathbb{E} [(a_i^T x)^2] = k.$$

12. Consider a multi-probe LSH setup with L hash tables, each using k hash functions. In multi-probe LSH, buckets that differ from the query bucket in only a few hash positions are more likely to contain near neighbors than those that differ in many positions.

To exploit this idea, we use perturbation vectors $\Delta \in \{-1, 0, +1\}^k$, and probe all buckets that are within Hamming distance at most s from the query bucket.

Under this strategy, the total number of distinct buckets probed is:

- A. $L \cdot k \cdot s$
- B. $L \cdot \sum_{n=1}^s k \cdot 2^n$
- C. $L \cdot \sum_{n=1}^s \binom{k}{n} \cdot 2^n$
- D. $L \cdot \sum_{n=1}^s k^n \cdot 2^n$

Solution Answer:(C) For distance n , choose n positions out of k in $\binom{k}{n}$ ways, and each can change in 2 ways (± 1), giving $\binom{k}{n} 2^n$ buckets.

Summing for $n \leq s$ and multiplying by L tables:

$$L \cdot \sum_{n=1}^s \binom{k}{n} 2^n$$

13. Let $m = 2 \times 10^6$, and consider the hash function

$$h_{a,b}(x) = ((ax + b) \bmod p) \bmod m,$$

where $a = 29$, $b = 51$, and $x \in [1, m]$.

When all integers in $[1, m]$ are hashed into m buckets, what is the length of the longest chain observed?

- A. 8
- B. 5
- C. 7
- D. 6

Solution Answer: (C) The hash function distributes integers in $[1, m]$ into m buckets. Since $a = 29$ and $b = 51$ define a linear congruential permutation modulo p , followed by modulo m , collisions arise due to the final reduction.

By implementation/observation, the elements are not perfectly uniformly distributed, and the maximum bucket load (longest chain) obtained is: 7

14. In a Johnson–Lindenstrauss (JL) transform, let $A \in \mathbb{R}^{k \times d}$ be a random Gaussian matrix whose entries are i.i.d. $N(0, 1/k)$. For any fixed vector $x \in \mathbb{R}^d$, what is the expected value of $\|Ax\|_2^2$?

- A. $\|x\|_2^2$
- B. $k \cdot \|x\|_2^2$
- C. $\|x\|_2$
- D. $\frac{1}{k}$

Solution Answer: (A) Each row of A has covariance $\frac{1}{k}I_d$. Hence,

$$\mathbb{E}[\|Ax\|_2^2] = \mathbb{E}\left[\sum_{i=1}^k (a_i^T x)^2\right]$$

For each row a_i ,

$$\mathbb{E}[(a_i^T x)^2] = \frac{1}{k} \|x\|_2^2$$

So,

$$\mathbb{E}[\|Ax\|_2^2] = k \cdot \frac{1}{k} \|x\|_2^2 = \|x\|_2^2$$

15. With $m = 3000$, $a = 45$, $b = 60$, hash integers in $[1, 4m]$. Longest chain is?

- A. 3
- B. 6
- C. 4
- D. 5

Solution Answer: (D). Chain length peaks at 5.

16. Why does the variance of the random variable representing the difference between projections, $f_i(q) - f_i(p)$, depend directly on the Euclidean distance $R_{pq} = \|p - q\|_2$?
- A. Because the random projection vectors v_i are drawn from an IID Gaussian distribution $N(0, I)$, making the projection difference a Gaussian distribution scaled by the vector distance.
 - B. Because the random uniform shift $b_i \in [0, w)$ alters the variance linearly.
 - C. Because the bucket width w forces the variance of the data coordinates to compress quadratically.
 - D. Because multi-probe LSH strictly requires a medium-to-high dimensionality between 100 and 1000 to stabilize variance.

Solution Answer:(A) We have

$$f_i(x) = v_i^\top x + b_i$$

so

$$f_i(q) - f_i(p) = v_i^\top (q - p)$$

Since $v_i \sim \mathcal{N}(0, I)$ (or appropriately scaled Gaussian), the random variable $v_i^\top (q - p)$ is Gaussian with mean 0 and variance:

$$\text{Var}(v_i^\top (q - p)) = \|q - p\|_2^2 = R_{pq}^2$$

Thus, the variance depends directly on the Euclidean distance because Gaussian projections preserve variance proportional to squared norm.

The shift b_i cancels out in the difference and does not affect variance.

17. Consider a Johnson–Lindenstrauss (JL) random projection matrix where each row contains only $O(1)$ non-zero entries (i.e., a very sparse embedding). Which of the following best explains why such a construction may fail to satisfy the JL guarantee?
- A. The transformation becomes non-linear, violating linearity assumptions.
 - B. There exist input directions that may be mapped to the zero vector or severely shrunk, violating distance preservation.
 - C. The embedding necessarily increases the output dimension beyond the original space.
 - D. The matrix fails only to be orthogonal, but still preserves distances up to scaling.

Solution Answer:(B) In a very sparse JL matrix, some coordinates may not be sampled at all in a row. Hence, for certain input vectors aligned with missing coordinates, the projection can lose all energy in those directions. This leads to failure in norm preservation required by JL.

18. Consider the Count-Min sketch where an estimate for a query x is typically obtained by taking the median (or minimum) over multiple hash tables.

Suppose instead that, for each query x , you compute the estimate using

$$\max_i A[i, h_i(x)]$$

i.e., you take the maximum over the k estimators instead of the usual aggregation rule.

Which of the following statements are true?

- A. Using the maximum breaks the theoretical guarantees of the Count-Min sketch.
- B. The guarantees of Count-Min still hold, but the estimates may become worse in practice.
- C. The resulting estimates are strictly better than those produced by the standard Count-Min sketch.
- D. None of the above.

Solution Answer:(A) Count-Min sketch guarantees rely on taking the **minimum** (or median) across hash tables to control overestimation due to collisions.

Taking the **maximum** instead amplifies collision noise and destroys the error bound guarantees. Hence, the Count-Min theoretical guarantees no longer hold.

19. Consider Euclidean Locality Sensitive Hashing (LSH), where data points are projected onto a random line and then discretized into buckets of fixed width w .

Which of the following best describes the construction of the hash function in Euclidean LSH?

- A. Sorting pairwise distances between data points
- B. Random projection of points followed by quantization into buckets
- C. Reconstruction of data using PCA components
- D. Clustering using graph-based methods

Solution Answer:(B) In Euclidean LSH, each data point is first projected onto a random vector (random line), and then the projection value is discretized into equal-width intervals (quantization into buckets). This ensures that nearby points have a high probability of falling into the same bucket.

20. In Euclidean LSH, a point $x \in \mathbb{R}^d$ is hashed using a random projection followed by bucketization with width $w = 2$. If the projected value of a point is $f(x) = 7.3$, into which bucket does it fall?
- A. 3
 - B. 4
 - C. 5
 - D. 6

Solution Answer:(A) Bucket index is given by:

$$\left\lfloor \frac{f(x)}{w} \right\rfloor = \left\lfloor \frac{7.3}{2} \right\rfloor = \lfloor 3.65 \rfloor = 3$$